

Final Documentation

Unified Military Analytics and Comparison Dashboard

Author: Yogeshwar Vadla

Internship: Infosys Springboard

Mentor: Sirisha Ma'am

Domain: Data Visualization

1. Methodology

The Unified Military Analytics and Comparison Dashboard was developed using a structured end-to-end data analytics pipeline consisting of:

1. Data Acquisition (Web Scraping)
2. Data Cleaning & Standardization
3. KPI Engineering
4. Dashboard Development in Power BI
5. Testing & Validation

The methodology ensures scalability, analytical reliability, and professional visualization standards.

1.1 Scraping Method

The data collection process was implemented using an automated Python-based scraping pipeline.

Data Source

- Publicly available military data from GlobalFirepower.com
- 145+ countries
- 50+ military, economic, and geographic metrics

Scraping Process

The scraping methodology consisted of:

Step 1: Base Data Extraction

- Scraped the countries listing page.
- Extracted:
 - Global Rank
 - Country Full Name
 - Country Short Name
 - Power Index (PwrIndx)

Step 2: Metric-Level Scraping

- 55+ metric-specific URLs were predefined.
- Each page was parsed using structured HTML DOM selectors.
- Extracted metrics included:
 - Manpower
 - Aircraft
 - Tanks
 - Naval Fleet
 - Defense Budget
 - Economic Indicators
 - Geographic Metrics

Step 3: Ethical Scraping Practices

- Used browser-like headers
- Implemented request throttling (time delays)
- Validated HTTP responses
- Skipped malformed values safely

Step 4: Raw Data Storage

- Data was stored without transformation
- Exported as:
 - CSV
 - Excel
- Served as input for cleaning phase

This approach ensured automation, reliability, and reproducibility.

1.2 Data Cleaning & Structuring

The raw dataset was transformed into an analytics-ready dataset.

Cleaning Steps

- Removed commas, currency symbols, unit labels
- Converted numeric columns to proper data types
- Standardized country identifiers
- Preserved one-row-per-country integrity
- Handled missing values safely
- Ensured compatibility with Power BI

Final cleaned dataset:

- 145 countries
- 50+ standardized numeric metrics
- Saved as military_cleaned.csv

1.3 KPI Definitions

Three strategic KPIs were engineered to enhance analytical depth.

KPI 1: Power Index Rank Gap

Definition:

Difference between computed rank (based on Power Index) and official reported rank.

Formula:

Computed Rank – Official Rank

Purpose:

Identifies countries that may be:

- Strategically underrated
- Strategically overrated

KPI 2: Assets per Capita

Definition:

Total military assets normalized by population.

Formula:

(Aircraft + Helicopters + Tanks + Naval Fleet) / Total Population

Purpose:

Provides fair comparison across countries with different population sizes.
Measures military density rather than raw volume.

KPI 3: Budget-to-GDP Ratio

Definition:

Defense budget relative to national economic capacity.

Formula:

Defense Budget (USD) / GDP (PPP)

Purpose:

Indicates how much economic capacity is allocated toward defense.

1.4 Metadata Enrichment

Additional geopolitical attributes were added:

- Continent
- Region
- NATO Membership Flag

This enabled:

- Alliance-level analysis
- Regional comparisons
- Strategic grouping

2. Power BI Dashboard Usage Guide

The final Power BI report file:

File Name: global_military_firepower_2025.pbix

The dashboard consists of six integrated pages:

1. Quick Stats
2. Nation Overview
3. Global Nations Economy
4. Global Military Insights
5. Compare Powers
6. Coalition Builder

2.1 How to Open the Dashboard

1. Install Microsoft Power BI Desktop.
2. Open Power BI Desktop.
3. Click **File → Open**.
4. Select:
global_military_firepower_2025.pbix
5. Wait for the data model to load.
6. Use navigation buttons to switch between pages.

2.2 How to Use Each Dashboard Page

A. Quick Stats

Purpose: Global-level military overview.

How to Use:

- Apply filter:
 - Country search
 - Select a country

- Observe:
 - Top 10 military powers
 - Defense budget KPIs
 - Manpower summaries
 - Geographic distribution

Use Case:

High-level executive overview.

B. Nation Overview

Purpose: Detailed country profile.

How to Use:

- Select a country from slicer.
- View dynamic KPIs:
 - Population
 - Defense Budget
 - Power Index Rank
 - Assets per Capita
- Analyze asset breakdown visuals.

Use Case:

Country-specific strategic assessment.

C. Global Nations Economy

Purpose:

Analyze economic strength and defense spending patterns across countries.

How to Use:

- Apply filters such as country.
- View economic KPIs:
 - Budget-to-GDP Ratio
 - Budget to GDP
 - Country Defense Budget
 - Assets per Capita
- Compare countries using Budget vs GDP visuals.
- Identify top defense spenders using ranking charts.
- Analyze regional spending distribution patterns.

Use Case:

Evaluate how economic capacity supports military expenditure and defense prioritization.

D. Global Military Insights

Purpose:

Macro-level analysis of global military strength across air, land, naval, geographic, and economic dimensions.

How to Use:

- Review the **Total Submarines** line chart to identify countries with strong underwater naval capabilities.
- Analyze the multi-metric comparison chart showing:
 - Total Military Helicopters
 - Tanks
 - Total Naval Fleet
 - Fighter Aircraft
- Examine the clustered bar chart for:
 - Total Military Aircraft
 - Total Military Helicopters
- Observe the horizontal bar chart for:
 - Total Land Area (sq. km)
- Study the pie chart:
 - Defense Budget (USD) by Region
- Use scroll bars to navigate and compare additional countries.

Use Case:

Strategic evaluation of global military capability distribution and identification of dominant nations across different military domains.

E. Compare Powers

Purpose: Side-by-side comparison of two countries.

How to Use:

1. Select Country A.
2. Select Country B.
3. Compare:
 - Manpower
 - Aircraft
 - Naval Fleet
 - Defense Budget

- KPIs

4. Observe visual differences across asset categories.

Use Case:

Bilateral military evaluation.

F. Coalition Builder

Purpose: Simulate multi-country alliance strength.

How to Use:

1. Select multiple countries in Coalition slicer.
2. Select a comparison country
3. View aggregated totals:
 - Total Manpower
 - Total Defense Budget
 - Total Aircraft
 - Total Naval Fleet
4. Compare coalition vs single country.

Use Case:

Alliance simulation and geopolitical analysis.

3. Testing & Validation Summary

Before final submission, the dashboard was tested for:

- Data accuracy validation
- Slicer functionality
- DAX measure correctness
- Navigation integrity
- Layout consistency
- Performance optimization
- Edge-case scenario stability

All dashboards were verified and debugged.

4. Conclusion

The Unified Military Analytics and Comparison Dashboard successfully integrates:

- Automated web scraping
- Data cleaning and standardization
- Strategic KPI engineering
- Advanced DAX-based comparison logic
- Multi-page interactive Power BI dashboards
- Coalition simulation capability

The project demonstrates a complete end-to-end data analytics workflow from data acquisition to professional BI deployment.

The final Power BI application is fully functional, interactive, validated, and ready for professional evaluation.